Introduction to Computer Vision (ECSE 415)
Assignment 2: Image Matching and Face Detection

1 Q3 Instruction

This dataset folder celeb_faces contains 400 178x218 images of celebrities. Load all the images and make a dataset in python. Convert each image to grayscale then flatten each image and append to a large array. This would give a large array of shape 400x38804 where each row is a flattened image.

Show the mean (average) face image of the training set. Calculate the eigenvectors for the matrix $D^T D$ where D is defined by subtracting the mean face from large array. Eigenvectos and eigenvalues can be found using online packages. Order the eigenvectors by descending order of eigenvalues. Print out the first four eigenvalues.